



COVID VACCINATION RATES IN THE US

When will the U.S. Population
Be Fully Vaccinated?

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RESEARCH QUESTION

When will 90% of the US population be fully vaccinated?

- Fully Vaccinated Meaning
 - Receives full dosage of a particular COVID-19 Vaccine
 - Pfizer
 - 2 doses
 - Moderna
 - 2 doses
 - Johnson & Johnson
 - 1 dose



WHY IS THIS IMPORTANT? ➔



- **Covid-19 Cases Statistics**

- As of October 22, 2021, there have been a total of 243 million Covid-19 Cases worldwide (Statista)
 - 45.1 million people have contracted Covid-19 alone in the US
- Around 4.9 million have died worldwide from Covid-19 (Statista)
 - 730,000 have died in the U.S. from Covid-19
- Currently 66% of the U.S. population is vaccinated with only 57% being fully vaccinated (The New York Times)

- **Covid-19 Importance**

- Direct Impact on Our Community
- Even those that recovered still can suffer from lasting health issues due to Covid-19 implications
 - long-term breathing problems
 - heart complications
 - chronic kidney impairment
- Adopting 90% full vaccination in the U.S. would help minimize death from Covid-19

HOW WILL THE DATA BE USED?



- Goal: Predict when will 90% of the U.S Population will be fully vaccinated
- Dataset: Gabriel Preda's "COVID-19 World Vaccination Progress" dataset on Kaggle, which is frequently updated with current numbers on total vaccinations.
- Focus on extracting data specifically tracking the progress within the United States

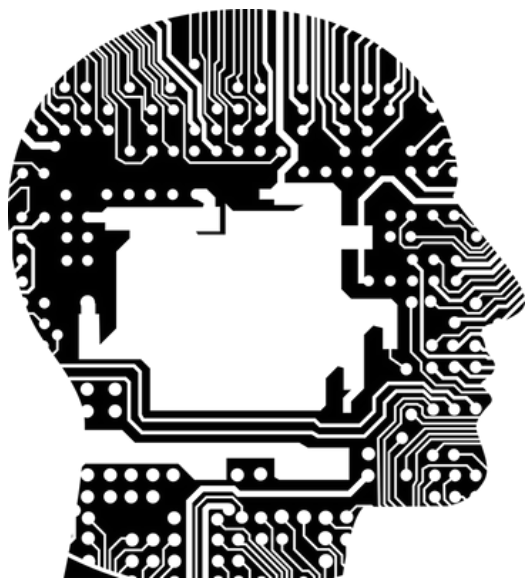
Note: We are focusing on the population that is eligible to receive the vaccine. Children under 12 are not included in the dataset and are not considered for our prediction.

MACHINE LEARNING ALGORITHM TO BE USED



- Linear Regression Model

- Supervised Machine Learning Model
- "...assumes a linear relationship between variables." (Jidge, Abhay.)
- "One variable is considered to be an explanatory variable, and the other is considered to be a dependent variable. (Yale.)"



SOURCES OF BIAS

- Individuals that cannot get the vaccine
 - Allergic to certain vaccine ingredients
 - People with certain medical conditions
- People affected by homelessness
 - According to the National Academy for State Health Policy, "...at least 20 states don't include people living in homeless shelters in their vaccine distribution plans..." (Pew Trusts)
- Accessibility/Proximity to Nearest Provider
- Issues with Processing the Data
 - Incomplete/missing data for certain days



AI IMPACT & ETHICAL IMPLICATIONS



- AI could be used in future COVID-19 endeavors to help determine statistics, such as vaccination rates.
- AI can be used internationally for COVID-based purposes
- The algorithm can be reused for similar statistics-based projects/observations.

DATASETS TO BE USED



- Kaggle: COVID-19 World Vaccination Progress - Gabriel Preda
- Website: Our World in Data
- Github: Our World in Data

Summary

- As COVID-19 continues to affect populations everywhere, effective vaccination rates are key to diminishing death tolls
- We can apply regression modeling techniques within Machine Learning to learn more about different issues -- not just those involving COVID-19
- We hope to see the direction in which our society is going in regards to combating the virus and learn more about the effect of COVID-19 on our population

SOURCES



- [Towards Data Science: Understanding Regression Using COVID-19 Dataset](#)
- [Towards Data Science: A Simple Guide to Linear Regression Using Python](#)
- [Towards Data Science: A Complete Guide to Linear Regression Analysis](#)
- [Yale: Linear Regression](#)
- [Pew: States Fail to Prioritize Homeless People For Vaccines](#)
- [USA Facts: COVID Vaccine Progress Tracker – Vaccinations by State](#)
- [NY Times: Interactive World COVID Vaccination Tracker](#)
- [Statista: COVID-19 Cases Worldwide, Recoveries, Deaths](#)